

Original article

Prenatal cytogenetic diagnosis study of 2782 cases of high-risk pregnant women

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Keywords: chorionic villus sampling; amniocentesis; umbilical cord blood collection; chromosomal karyotype; prenatal diagnosis

Background Prenatal diagnoses are extremely advantageous for pregnant women with high-risk indicators and can help prevent the birth of malformed infants. However, no large-scale statistical study analyzing the correlation between fetal chromosome disorders and abnormal indicators during pregnancy has been done in China. The objectives of this study were to diagnose and analyze fetal chromosome abnormalities, determine the feasibility of the various prenatal test methods and establish diagnostic guidelines for the early, middle, and late trimesters.

Methods From January 2004 to May 2009, 2782 pregnant women at high-risk underwent prenatal diagnoses. Categorized data expressed as either actual counts or percentages were analyzed by the chi-square or Fisher's exact test. Chorionic villus sampling was performed in the early-trimester (10–12 weeks of gestation), amniocentesis in mid-trimester (16–28 weeks of gestation), and umbilical cord blood collection in mid- or late-trimester (16–37 weeks of gestation). In 51 cases either autopsy samples from intrauterine fetal deaths or placental tissues from aborted fetuses were tested.

Results Chromosomal abnormalities were observed in 3.99% (111/2782) of the samples. Overall, the success rate of cytogenetic analysis for high-risk pregnancy groups was 98.17% (2731/2782). It was significantly less successful when used to analyze data from the chorionic villus sampling compared with that from amniocentesis and umbilical cord blood ($P=0.000$). Abnormal chromosome carriers had the highest percentage of abnormal chromosomes (67.86%) when compared with chromosomal abnormalities in patients with ultra-sonographic “soft markers” (11.81%), advanced maternal age (4.51%) and those who had positive serum screening results ($P=0.000$).

Conclusions Invasive prenatal diagnostic techniques are feasible tools for confirming fetal chromosomal abnormalities. Abnormal chromosomes detected in one of the parents carrying abnormal chromosome, ultrasound soft markers, advanced maternal age or positive serum screening results were associated with a higher frequency of fetal genetic diseases.

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With the advancement of prenatal diagnostic techniques, more fetal abnormalities can be detected. The incidence of chromosomal disorders is about 0.5% in live newborns.^{1,2} Because there are no effective treatments for fetal abnormalities, prenatal diagnoses are important means for women to diagnose abnormalities in the unborn child. Chorionic villus, amniocytes or umbilical blood cell sampling during pregnancy provide reliable methods for performing prenatal diagnoses. The present study examined 2782 cases of high-risk pregnancies using prenatal diagnostic techniques. Using this information we were able to summarize the correlation between fetal chromosome abnormalities and various abnormal indicators during pregnancy. Reliable criteria for early, middle, and late trimester diagnoses were established thus increasing the detection rate of prenatal fetal chromosomal disorders.

METHODS

Subjects

From January 2004 to May 2009, there were 2782 cases of pregnant women with various high-risk indicators in

the Prenatal Diagnosis Center of Peking University People's Hospital. Among those, there were 152 cases of chorionic villus sampling (CVS), 2346 cases of amniocentesis (AMC), 233 cases of umbilical blood collection (UBC), and 51 cases of tissues collected from intrauterine fetal deaths or aborted fetuses. The median age of the mothers was 32.75 years (range 20–44 years with 44.64% women over 35 years old). The median gestational week was 20.81 weeks (range 10–37 weeks). Prenatal sampling criteria were as follows: positive serum screening results for Down syndrome and other aneuploidies, advanced maternal age (AMA), a history of intrauterine fetal death, aborted fetuses or a family history of genetic disease with one of the couple carrying abnormal chromosomes, a history of contact with

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teratogens or medications during pregnancy, or abnormal ultrasonographic indicators, and small for gestational age (SGA) provided there are other abnormal ultrasonographic markers and/or positive serum screening results. Pregnant women without any abnormal indicators who asked for examination of the fetal chromosomes were also enrolled. A total of 2482 subjects fulfilled one criterion, 277 subjects fulfilled two criteria and 23 subjects fulfilled three criteria simultaneously. Table 1 lists the prenatal screening criteria.

Table 1. Prenatal diagnostic criteria

Prenatal diagnostic criteria	Case number (n (%))
AMA (greater than or equal to 35 years)	1242 (44.64)
Positive serum screening results for Down syndrome	1153 (41.45)
Positive serum screening results for trisomy 18	87 (3.13)
One of the couple carrying abnormal chromosomes	28 (1.01)
A history of intrauterine fetal death or aborted fetuses	77 (2.77)
Ultrasonographic abnormal indications	288 (10.35)
A history of contact with teratogens	79 (2.84)
Taking medication during early-trimester	56 (2.01)
A family history of genetic disease	37 (1.33)
Else	40 (1.44)
Asking for examination of the fetal chromosomes without any indications	18 (0.65)

Chorionic villus sampling, amniocentesis and fetal umbilical cord vein collection were done with ultrasound monitoring and with a signed informed consent from each patient. Informed consents were reviewed and approved by the local Ethics Committee of Peking University People's Hospital.

Chorionic villus sampling (CVS)

CVS were performed between 10 and 12 weeks of gestation. About 8–10 mg (wet weight) samples of CVS were obtained transabdominally. Maternal decidua was removed and CVS was dissected under an anatomical microscope and dispersed into single cells. The cells were cultured in culture media (Gibco, USA and Israel) and grew in an incubator for 9–10 days at 37°C and 5% CO₂. Cell growth was monitored every day. The cells were collected when multiple clones with numerous metaphase cells were observed under an inverted microscope. G banding (plus C-banding when necessary) was performed for chromosomal sample preparation, following the ISCN (1985) standard to analyze the karyotypes.

Amniocentesis (AMC)

AMC was performed between 16 and 28 weeks of gestation. Twenty milliliter of amniotic fluid was collected (the first 1–2 ml of amniotic fluid was discarded) and centrifuged at 1000 r/min for 10 minutes. After discarding the supernatant, 2.5–3.0 ml of cell suspension was inoculated in 5 ml of culture media (Gibco, USA and Israel) and cultured for 10–11 days at 37°C and 5% CO₂. The cells were then treated using the same procedure as described for CVS chromosomal analyses.

Fetal umbilical cord vein blood collection (UBC)

Fetal blood was collected from the umbilical vein

between 16 and 37 weeks of gestation. The needle lumen and the syringe had been rinsed with heparin before the procedure to avoid blood clotting; 2 ml of cord blood was aspirated and the presence of fetal blood without maternal blood contamination was confirmed by the hemoglobin patterns and Kleihauer's test.³ The blood was inoculated in 5 ml of 1640 culture media (1640 media, China) and cultured for 72 hours at 37°C. The cells were then harvested, and chromosomal samples were prepared following the same procedures used for the CVS and AMC samples.

Examination of fetal tissues after intrauterine fetal death or abortion

Tissue samples from autopsies performed after intrauterine death or placental fetal tissues obtained after abortion were cultured as described above for CVS. Chromosomal preparation and karyotypes were analyzed following the procedures described for CVS.

Screening protocols for Down syndrome and other aneuploidies⁴⁻⁷

The risk factors for Down syndrome and other aneuploidies were assessed based on combined screening indicators for the early and middle trimesters. For the early trimester (8–13⁺₆ gestation weeks), these indicators included nuchal translucency (NT) thickness in combination with measurement of pregnancy-associated plasma protein A and human chorionic gonadotropin (hCG), gestation week, age and weight of the pregnant women. For the middle trimester (15–20 gestation weeks), maternal serum alpha-fetoprotein (AFP), β -hCG, and unconjugated estriol (uE3) were measured and used in combination with the gestation week, age and weight of the pregnant women. Perkin Elmer Precisely prenatal diagnostic software was used to combine the early and middle trimester indicators and calculate the risk of Down syndrome and other aneuploidies (high risk: >1:270; low risk: <1:270).

Statistical analysis

The categorical data were expressed as counts and percentages. Differences among groups for categorical data were evaluated by using chi-square test and Fisher's exact test was used when expected value was less 5 for the fourfold table data. A two-tailed $P < 0.05$ was considered to indicate statistical significance. Statistical analyses were performed using SAS 9.0 software (SAS Institute Inc., USA).

RESULTS

Abnormal karyotypes detected in pregnant women with high-risk indicators

A total of 2782 high-risk pregnant women were studied during over a 5-year period from January 2004 to May 2009. Within this group of test patients, the success rates of cytogenetic analyses were 98.17% (2731/2782). Chromosomal abnormalities were observed in 3.99%

Table 2. Abnormal chromosome karyotype of 9 fetuses via CVS

No.	Age (years)	Gestational weeks	Prenatal diagnostic criteria	Chromosome karyotype of fetus	Outcomes of the pregnancy
1	37	10	A history of intrauterine fetal death or abortion	45,X	Termination
2	41	11	Advanced maternal age	45,X	Termination
3	39	10*	A history intrauterine fetal death or abortion	69,XXY	Termination
4	37	12	Rest abortion	47,XY,+22	Termination
5	39	11*	A history of fetus abnormality	45,X/46,XX	Termination
6	37	11*	Rest abortion	47,XX,+16	Termination
7	39	12*	Rest abortion	47,XX,+13	Termination
8	31	10*	A history of fetus abnormality	47,XXX	Termination
9	33	11	A history of fetus of congenital heart disease	47,XXY	Termination

(111/2782). Of these, trisomy 21 was established in 61 of 111 (54.96%), sex chromosomal abnormalities in 17 of 111 (15.32%), triploid in 5 of 111 (4.50%), and other chromosomal structural abnormalities in 28 of 111 (25.23%).

Chorionic villus sampling

Chromosomal abnormalities were observed in 9 of 152 CVS cases (5.92%). Within this group, trisomy 21 was detected in 3 of 9, sex chromosomal abnormalities in 5 of 9, and triploidy in 1 of 9. AMA patients accounted for the majority of those with chromosomal abnormalities (77.78%, Table 2).

Amniocentesis

Chromosomal abnormalities were observed in 53 of 2346 AMC samples (2.26%). Of these, trisomy 21 was established in 26 of 53 (49.06%), sex chromosomal abnormalities in 5 of 53 (9.43%), triploidy in 1 of 53 (1.89%), and other chromosomal structural abnormalities in 21 of 53 (39.62%). The distribution of patients was as follows: AMA patients (37.74%), patients with high risk for Down syndrome (32.08%), abnormal chromosome carriers (30.19%), and patients with abnormal ultrasound markers (26.42%) ($P < 0.05$, Table 3).

Fetal umbilical cord vein blood collection

Chromosomal abnormalities were observed in 28 of 233 (12.02%). Of these, trisomy 21 was detected in 24 of 233 (85.71%), sex chromosomal abnormalities in 2 of 233 (7.14%), triploidy in 1 of 233 (3.57%), and other chromosomal structural abnormalities in 1 of 233 (3.57%). Patients with abnormal ultrasound markers accounted for the majority of those patients (78.57%), followed by AMA (50%), abnormal chromosome carriers (10.71%), and patients with positive serum screening results (7.14%) ($P < 0.01$ between two groups, Table 4).

Tissue culture samples from spontaneous abortion

Tissue samples from spontaneous abortions were obtained with permission of the patient. Chromosomal abnormalities were observed in 21 of 51 abortion samples (41.18%). In the group of those 21 samples trisomy 21 was detected in 9 of 21 (42.86%), triploidy in 2 of 21 (9.52%), sex chromosomal abnormalities in 5 of 21 (23.81%), and other chromosomal structural abnormalities in 5 of 21 (23.81%). AMA accounted for 57.14% of all abnormal cases, while patients <35 years old accounted for 42.86%.

Correlation between AMA and abnormal fetal chromosomes

Among 2782 pregnant women in this study, there were 1242 in the AMA group (>35 years). This was approximately 44.64% of the total cases. The group was subdivided into AMA patients without serum screening (907), those with positive serum screening results (254) and low-risk AMA patients who requested an examination of the fetal chromosomes (81). Chromosomal abnormalities were observed in 56 of 1242 (4.51%) in AMA and 55 of 1540 (3.57%) in small for gestational age (SGA) group (females <35 years). There was no statistically significant difference between AMA and SGA with other abnormal markers ($P = 0.209$). Among 1242 AMA cases, chromosomal abnormalities were observed in 36 of 907 AMA without serum screening samples (3.97%), 19 of 254 (7.48%) AMA samples with positive maternal serum screening results, and 1 out of 81 (1.23%) low-risk AMA patients requesting an examination of the fetal chromosomes.

Correlation between positive serum screening results and abnormal fetal chromosomes

Among the 1153 pregnant women with positive serum screening results for Down syndrome and other aneuploidies, chromosomal abnormalities were observed in 16 of 1153 cases (1.39%).

Correlation between abnormal ultrasonographic indicators and abnormal fetal chromosomes

There were 288 cases with various abnormal ultrasonographic indicators that included the following abnormalities — bilateral or unilateral choroid plexus cysts, nuchal translucency thickening, hydrocephalus, fetal hydrops, abnormal limbs, intracardiac focus, absence of megalblase, and fetal organ deformities such as polycystic kidney, pelviectasis, hydronephrosis, widened lateral ventricles, cardiac abnormalities, megacolon, duodenal stenosis, esophageal stenosis, and internal organ ectropion. Chromosomal abnormalities were observed in 34 of 288 (11.81%).⁸

Correlation between parental abnormal chromosome carriers and abnormal fetal chromosomes

Among 28 cases with either of the parents carrying abnormal chromosomes, chromosomal abnormalities were observed in 19 of 28 (67.86%). Triplets were counted as 3 separate cases.

Table 3. Abnormal chromosome karyotype of 53 fetuses via AMC

No.	Age (years)	Gestational weeks	Prenatal diagnostic criteria	Chromosome karyotype of fetus	Outcomes of the pregnancy
1	23	19 ⁺	NT=0.7	47,XX,+21	Termination
2	27	20	High risk of Down syndrome screening results	47,XY,+21	Termination
3	40	17	A history of delivering a Down syndrome child	47,XY,+21	Termination
4	33	18	Fetus with abnormal heart indications on ultrasound examination	47,XX,+21	Termination
5	39	17 ⁺	Advanced maternal age	47,XX,+21	Termination
6	36	18	Advanced maternal age	47,XY,+21	Termination
7	33	20	Fetus with dilative kidney or hearts abnormal indications on ultrasound examination	47,XY,+21	Termination
8	33	19	High risk of Down syndrome screening results	47,XY,+21	Termination
9	37	21	High risk of Down syndrome screening results	47,XX,+21	Termination
10	34	18	High risk of Down syndrome screening results	47,XX,+21	Termination
11	27	19	High risk of Down syndrome screening results	47,XY,+21	Termination
12	36	18 ⁻	Advanced maternal age	47,XX,+21	Termination
13	23	21	High risk of Down syndrome screening results	47,XY,+21	Termination
14	35	23 ⁺³	High risk of Down syndrome screening results	47,XY,+21	Termination
15	27	19	NT=0.3	47,XY,+21	Termination
16	37	19	A history of delivering a Down syndrome child	47,XX,+21	Termination
17	36	24 ⁺⁶	NF=1.0, fetus with unilateral choroid plexus cysts	47,XY,+21	Termination
18	37	24	Fetus with F4 syndrome	47,XX,+21	Termination
19	33	21 ⁺	High risk of Down syndrome screening results, NT=0.4	47,XY,+21	Termination
20	30	19 ⁺	The chromosome of husband: 46,XY,21p ⁻	46,XX,21p ⁻	Continuing
21	36	20	Fetus with bilateral choroid plexus cysts	47,XX,+18	Termination
22	28	19	Fetus with unilateral choroid plexus cysts, high risk of trisomy 18 syndrome screening results	47,XX,+18	Termination
23	33	22	Fetus with unilateral choroid plexus cysts	47,XX,+18	Termination
24	32	19 ⁺	High risk of trisomy 18 syndrome screening results	47,XX,+18	Termination
25	33	22	High risk of Down syndrome screening results	47,XY,+18	Termination
26	36	21	High risk of Down syndrome screening results	47,XY,+22/46,XY	Termination
27	39	18	Advanced maternal age	47,XY,+22	Termination
28	26	16 ⁺	Fetus edema	45,X	Termination
29	32	22 ⁺	A history failed pregnancy or abortion	47,XXY	Termination
30	35	18	Advanced maternal age	47,XXY	Termination
31	36	18	High risk of Down syndrome screening test	47,XXY	Termination
32	38	20	Advanced maternal age	45,X/46,X,i(X)(q10;q10)	Termination
33	30	18	A history of abortions, the chromosome of husband: 45,XY,t(13;14)(q10;q10)	45,XY,t(13;14)(q10;q10)	Continuing
34	22	18	The chromosome of husband: 46,XY,t(13;15), IVF	The chromosome of the triplets: 46,XY,t(13,15)	Continuing
35	36	18	Advanced maternal age, a history failed pregnancy or abortion; the chromosome of the prenatal woman: 46,XX,t(13;14)	46,XY,t(13;14)(q10;q10)	Continuing
36	37	20	The chromosome of the pregnant woman: 46,XX,t(Y;14)(Yq12;14p11)	46,XY,t(Y;14)(Yq12;14p11)	Continuing
37	34	19	A history of delivering trisomy 21; the chromosome of husband: 46,XY,t(Y;15)	46,XX,t(Y;15)(Y12;15qter)	Continuing
38	36	17 ⁺	Advanced maternal age, the chromosome of husband: 46,XY,t(Y;13)	46,XY,t(Y;13)(Yq12;13p11)	Continuing
39	35	21	Advanced maternal age, the chromosome of the prenatal woman: 46,XX,t(Y;15)	46,XY,t(Y;15)(Y12;15qter)	Continuing
40	27	18	Four times abortions. The fetus with multiple deformities. The chromosome of husband: 46,XY,t(3;18)(q22;p11)	46,XY,-18,+t(3;18)(q22;p11)	Termination
41	37	19	Oligohydramnios, fetal growth restriction	69,XXX	Termination
42	34	22 ⁺	High risk of Down syndrome screening results	46,XX,inv(9)(p11;q13)	Continuing
43	34	19	High risk of Down syndrome screening results	46,XX,inv(9)(p11;q13)	Continuing
44	34	19 ⁺	High risk of trisomy 18 screening results, the chromosome of husband: 46,XY, inv(9)(p11;q13)	46,XY,inv(9)(p11;q13)	Continuing
45	41	19	Fetus with bilateral choroid plexus cysts	46,XX,inv(9)(q11;q22)	Continuing
46	27	20	The chromosome of husband: 46,XY,inv(9)(p13;q22)	46,XY,inv(9)(p13;q22)	Continuing
47	36	19	A history intrauterine fetal death or abortion, advanced maternal age	46,XY,inv(9)(p21;q13)	Continuing
48	36	19	The chromosome of husband: 46,XY,(1qh+)(Y>18)	46,XX,(1qh+)	Continuing
49	28	19	The chromosome of husband: 46,XY,t(19;22)(p11;q13)	46,XY,t(19;22)(p11;q13)	Continuing
50	33	22	The chromosome of husband: 46,XY,t(6;14)(q25;q13)	46,XY,t(6;14)(q25;q13)	Continuing
51	35	21	High risk of trisomy 18 screening results, the chromosome of the pregnant woman: 46,XX,t(2;6)(p11;q11)	46,XY,t(2;6)(p11;q11)	Continuing

F4: Tetralogy of Fallot. IVF: *in-vitro* fertilization. inv: inversion. t: translocation. der: derivative chromosome.

Table 4. Abnormal chromosome karyotype of 28 fetus via UCB

No.	Age (years)	Gestational weeks	Prenatal diagnostic criteria	Chromosome karyotype of fetus	Outcomes of the pregnancy
1	35	29	Fetus with duodenal atresia, limb shortness	47,XX,+21	Termination
2	37	21	Fetus with ventricular septal defect, hydrocephalus	47,XX,+21	Termination
3	37	24	Endocardial cushion defects	47,XX,+21	Termination
4	33	24	NT=0.6, high Down syndrome screening results	47,XY,+21	Termination
5	39	31 ⁺	High Down syndrome screening results	47,XY,+21	Termination
6	39	25 ⁺	Fetus with intracardiac echogenic focus	47,XX,+21	Termination
7	42	31	Fetus with ventricular septal defect, aorta saddle	47,XX,+21	Termination
8	41	34	Fetus with VSD and ASD	47,XY,+21	Termination
9	40	28	Advanced maternal age and having a history of delivering Down syndrome child	47,XY,+21	Termination
10	22	29 ⁺	Fetus with VSD and cleft palate	47,XY,+21	Termination
11	28	27 ⁺	Endocardial cushion defects	47,XX,+21	Termination
12	31	33 ⁺	Fetus with thorax hydrops	47,XX,+21	Termination
13	34	25	The chromosome of the pregnant woman: 45,XX,t(21;22)	46,XX,t(21;22)	Termination
14	37	26	The chromosome of the pregnant woman: 45,XX,t(21;22)	45,XX,t(21;22)	Continuing
15	28	29	Fetus with F4	46,XY,t(15;21)	Termination
16	38	20 ⁺	Fetus with multiple deformities	47,XY,+18	Termination
17	24	22	Fetus with multiple deformities	47,XX,+18	Termination
18	24	18	Fetus with cleft palate, VSD and ventricle defect	47,XX,+18	Termination
19	35	30 ⁺	Esophageal atresia and VSD	47,XX,+18	Termination
20	27	22 ⁺	Fetus with bilateral choroid plexus cysts, VSD, single umbilical artery, artery agenesis	47,XX,+18	Termination
21	33	28	Fetus with multiple deformities	47,XX,+18	Termination
22	37	24	Fetus with multiple deformities	47,XX,+13	Termination
23	36	23 ⁺	Fetus with single ventricle and single atria	47,XY,+13	Termination
24	23	29 ⁺	Fetus with multiple deformities	47,XX,+13	Termination
25	42	28	Advanced maternal age	47,XXX	Termination
26	31	24	Fetus with pulmonary artery straitness and endaoerty dilatation, VSD	47,XXY	Termination
27	29	32	Fetal growth restriction, renal agensis, heart abnormality, oligohyramnios	69,XXX	Termination
28	31	28 ⁺	The chromosome of husband: 46,XY,inv(9)(p13;q13)	46,XY,inv(9)(p13;q13)	Continuing

VSD: ventricular septal defect. ASD: atrial septal defect.

Table 5. Abnormal chromosome karyotype of 21 aborted fetuses by tissue analysis

No.	Age (years)	Gestational weeks	Chromosome karyotype of fetus
1	27	12	47,XX,+13
2	32	14	47,XY,+13
3	39	15	47,XY,+13
4	35	14	47,XX,+16
5	47	16	47,XY,+16
6	43	17	47,XY,+16
7	36	18	47,XX,+18
8	39	17 ⁺	47,XY,+18
9	37	21	47,XX,+21
10	35	15 ⁺	45,XY,t(13;14)
11	36	14	45,XX,t(14;15)
12	36	17	45,X
13	33	18 ⁻	45,X
14	38	17 ⁺	45,X/46,X,i(X)(q10;q10)
15	34	16	45,X/46,XX
16	33	18	47,XXY
17	27	17	45,XY,del(13)(q23-ter)
18	32	19	69,XXX
19	36	18	69,XXY
20	27	14 ⁺	45,XX,-3,-18,+t(3;18)(q22;p11)
21	28	16	45,XY,-3,-18,+t(3;18)(q22;p11)

In summary there were 19 chromosomal abnormalities (67.86%) in 28 cases of abnormal chromosome carriers, 34 (11.81%) in 288 cases with ultrasonographic “soft markers”, and 56 in 1242 AMA (4.51%) and 19 in 1153 (1.65%) cases of positive screening results. Pregnant women or their partners carrying abnormal chromosomes had the highest rate of fetal chromosome abnormalities,

followed by those having various abnormal ultrasonographic indicators, AMA, and those with positive screening results ($P=0.000$ between two groups).

Outcome of pregnancy

Among 152 CVS sample cases, 139 cases were successfully diagnosed and the success rate of cytogenetic analyses was 91.45%. Among the successfully diagnosed cases, 9 cases of chromosomal abnormalities resulted in termination of the pregnancy. Among the remaining 130 cases it was advised to continue the pregnancy. Within that group there were three cases of fetal loss after the procedure and two of these had chromosomal abnormalities. Overall, the CVS-procedure-related fetal loss rate was 0.66%. Among 2346 of AMC, 2318 cases were successfully diagnosed, giving a 98.81% success rate. Among the successful cases, there were 33 cases of chromosomal abnormalities that terminated the pregnancy. Among the remaining 2285 cases, there were three cases of fetal loss. The AMC-related fetal loss rate was 0.13%. All 233 cases of UCB diagnoses were successful, giving a 100% success rate with this technique. Among the 28 cases of chromosomal abnormalities, it was advised that 26 women terminate the pregnancy. In the remaining two cases, there was one case of fetal loss. Overall, the UCB-related fetal loss rate was 0.43%. Follow-up examinations showed that the infants were all normally

developed. Among 51 cases of aborted fetus samples, 41 cases were successfully diagnosed yielding a success rate for cytogenetic analyses of 80.39%. There was no statistically significant difference in pregnancy loss rates between various invasive prenatal diagnostic techniques ($P=0.219$).

DISCUSSION

Rapid and safe prenatal diagnoses have become the standard of care in high-risk pregnancy. Rigorous judgment of various prenatal diagnostic results and timely termination of aberrant pregnancies is the key for allowing the patients to decide on the outcome of their pregnancies. In this study, 111 cases of fetal chromosomal abnormalities were detected and clinically managed. The present study summarized the correlation between fetal chromosome disorders and various high-risk indicators through the analyses of 2782 cases of prenatal diagnoses.

CVS is used routinely as a first-trimester diagnostic procedure for fetal genetic disease. The success of the procedure is dependent on the skill and experience of the medical personnel performing the technique. Its major advantage over AMC is that the sampling period is earlier. Its major flaw is the possibility that the technique can cause abortion and placental mosaicism. There have been concerns that CVS leads to limb reduction defects. Current data suggests that when the CVS is performed between 10 and 13 weeks of gestation, there is no significant difference in development of fetal defects compared with the incidence in the general population. CVS performed before 10 weeks of gestation increases the risk of limb reduction defects from 1% to 2%. We performed CVS at 10–12 gestation weeks. Among 152 cases, CVS-related fetal loss rate was identical to normal pregnancies (0.66%). There was no placental mosaicism and no limb reduction defects. CVS done in early pregnancy appears to be a relatively safe and reliable method of prenatal genetic diagnosis. If there is the possibility of a rare chromosome abnormality determined from the CVS results, any further diagnosis should be based on the later results from AMC.^{9,10}

AMC involves collecting amniotic fluid using ultrasound monitoring and analyzing fetal chromosomes, with relatively low risk to both the mother and the fetus. The major disadvantage of AMC is that the result is usually not available until after 18 weeks of gestation. Early AMC before 14 weeks has been shown to be associated with a significantly higher fetal loss rate and should not be used during that early time period. In the study, 99% of AMC samplings were performed between 16–22 gestation weeks. In our study there was one case of procedure, induced contractions and spontaneous abortion 24 hours after the procedure, but we believe that an intrauterine infection was the underlying cause. There were 2 cases of intrauterine fetal death at 2–3 weeks post-procedure, but the reasons are unknown at this time.

The remaining 2318 cases all had normal pregnancies. Overall, the AMC-related pregnancy loss rate was 0.13%. The success rate of cytogenetic analyses was 98.81% for the AMC sampling. AMC is a second-trimester prenatal diagnostic method and appears to be the safest method of prenatal diagnosis.^{11–13}

UCB is appropriate from 16 gestation weeks up to delivery. However, it is more difficult and has higher failure rate than AMC. It is crucial to verify that the blood sample is fetal, rather than maternal, before culturing it. From January 2004 to May 2009, we had 233 cases of UCB. There was one case of fetal loss. The UCB procedural-related pregnancy loss rate was 0.43% while the remaining 232 cases all had normal pregnancies. To reduce post-procedure complications, 99% of UCB sampling in our study was carried out between 24 and 37 gestation weeks. If diagnoses by CVS or AMC are unsuccessful, UCB is a fairly reliable, alternate diagnostic method.¹⁴

The cytogenetic success rate was 80.39% (41/51) from the spontaneous abortion samples. Approximately 50%–60% of the spontaneous abortions had fetal chromosome abnormalities.¹⁵ Genetic analyses of abortion samples could be offered to analyze the reasons for the abortion.¹⁶

Diagnoses for fetal chromosomal abnormalities are an essential part of prenatal care. Among 1153 cases from positive serum screening results, chromosomal abnormalities were observed in 19 of 1153 (1.65%) and accounted for 17.12% (19/111) of total abnormalities in the study. Fifteen cases with high risk for Down syndrome were observed: trisomy 21 in 10 of 15, trisomy 18 in 1 of 15, trisomy 22 in 1 of 15, 47 XXY in 1 of 15, and inv(9) in 2 of 15. Another 4 cases with high-risk for trisomy 18 were observed: trisomy 18 in 2 of 4, t(2:6) in 1 of 4 and inv(9) in 1 of 4. Positive serum screening results could not only suggest trisomy syndrome, but also other chromosome abnormalities.¹⁷ The use of the screening methods has enabled prenatal karyotyping to become cost effective.¹⁸

In AMA, the probability of chromosomal nondisjunction is increased, resulting in a higher risk for aneuploidy such as trisomy 21, trisomy 18 and trisomy 13. In this study, AMA was about 44.64% and SGA about 55.36% of the total cases. Chromosomal abnormalities were observed in 4.51% of AMA and 3.57% of SGA. There was no statistically significant difference between AMA and SGA with other abnormal markers ($P=0.209$). Among 1242 AMA cases, chromosomal abnormalities were observed in 3.97% for AMA without serum screening, 7.48% for AMA with positive serum screening results and 1.23% for AMA asking for examination with low risk. Cytogenetic analysis was useful in AMA with positive serum screening results ($P < 0.01$). We believe the serum screening risk limit should be specifically limited to the

population distribution of maternal age. Thus, prenatal diagnosis could be directly applied to AMA without serum screening.^{19,20}

If either of the parents carried an abnormal chromosome, there was a much higher probability of abnormal fetal chromosomes. In the present study, chromosomal abnormalities were observed in 19 out of 28 (67.86%) in which one of the parents carried abnormal chromosomes. Triplets were counted as three cases. In two cases partial chromosomal trisomy with fetal multiple deformities occurred and led to a terminated pregnancy. In the other 17 cases in which the fetal chromosomes were the same as one of the parents (a balanced translocation or an inversion chromosome) the pregnancy continued. This suggests that in families with a high risk of fetal chromosomal abnormalities prenatal diagnoses could effectively prevent the subsequent delivery of abnormal infants.

Ultrasound screening for chromosome abnormalities in gestation is highly sensitive. Some ultrasonographic indicators such as polyhydramnios, oligohydramnios or fetal structural abnormalities are indicative of fetal chromosomal abnormalities.²¹⁻²⁶ These "soft markers" might change with gestational week but had clinical significance throughout the whole pregnancy. There were 288 cases with abnormal ultrasonographic indications in the study. Of these, chromosomal abnormalities were observed in 34 of 288 (11.81%). Thus, if ultrasound results indicated an abnormal fetus, it should be recommended that patients undergo prenatal diagnoses to exclude the possibility of fetal chromosomal abnormalities.

In 2782 cases, chromosomal abnormalities were observed in 111 of 2782 (3.99%), which was higher than the reported rate of 1%–2%. This was most likely due to a high percentage of AMA patients and abnormal chromosome carriers. In the successfully diagnosed samples, pregnant women or their partners carrying abnormal chromosomes had the highest rate of fetal chromosome abnormalities, followed by those having various abnormal ultrasonographic indicators, AMA, and those with positive screening results ($P=0.000$ between two groups). The procedure-related fetal loss rate is one of the most important factors governing women's choice of an invasive prenatal test. In this study, early-trimester CVS, mid-trimester AMC, late-trimester UCB, there were no statistically significant differences among pregnancy loss rates in patients who underwent various invasive prenatal diagnostic methods ($P=0.219$). In summary, prenatal diagnoses should be performed in actual clinical situations to help determine if the birth of fetuses with abnormal chromosomes is best for both the parents and the unborn child. The discovery of an abnormality provides the option of termination or continuation of the pregnancy. The information obtained in the present study constitutes a basis for genetic counseling.²⁷

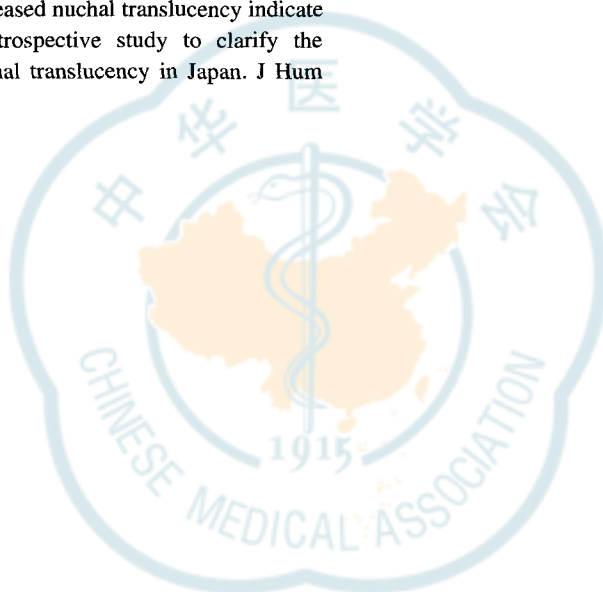
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